Theory and Practice of Security & Privacy Part 5

Security of Data

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Notice:
Security of System
<--> Security of Data

if System is unsecure => then Data on it is unsecure

general security concerns on System ... see previous course : Part 4

specific concerns on "Security of Data" ... this course : Part 5

Reminder

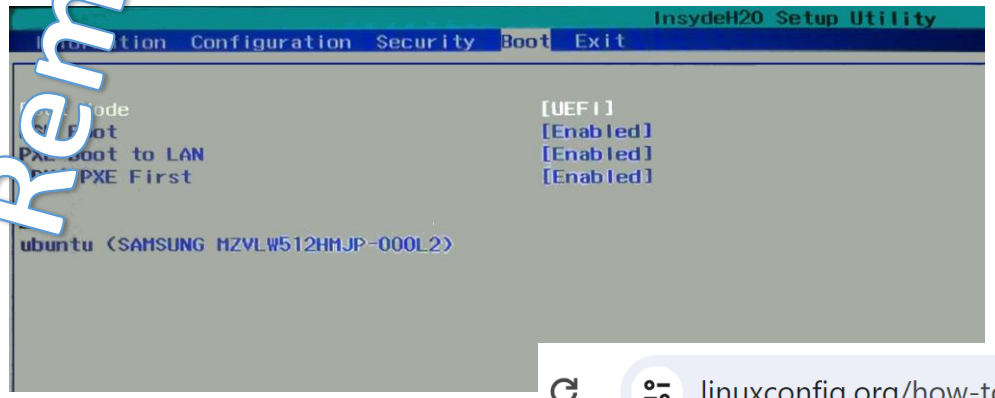
Access to PC Hard Disk
=> can read/write 010010101

Rule # 1 = Encrypt Disks

encryption key (in BIOS ? should be secured)

Reminder

Un-Encrypted Disk?
can we (do you need to) Boot ?





linuxconfig.org/how-to-mount-partition-with-ntfs-file-system-and-read-write-access

```
$ sudo mount -t ntfs /dev/sdb /mnt/ntfs
```



Un-encrypted Disk : risk for Data stolen
Anecdote as of 2024-02



REUTERS® World Business Markets Sustainability Legal Breakingviews Technology Investigations

Sports

Bag with Paris 2024 data lost, possibly stolen, prosecutor says

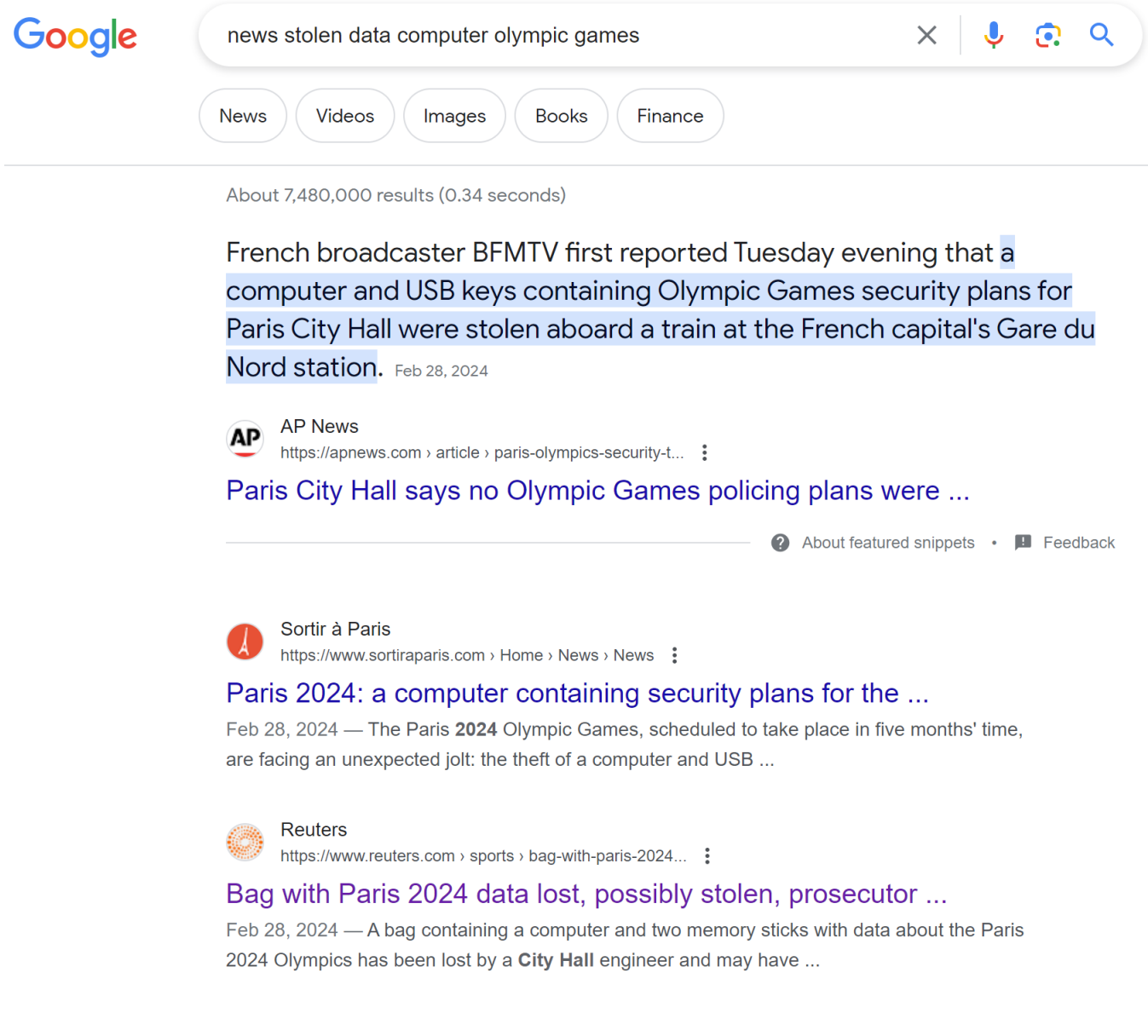
By **Juliette Jabkhiro** and **Julien Pretot**

February 28, 2024 2:36 PM GMT+1 · Updated 10 days ago

Aa

"if" stolen Pc (Disk) was encrypted => "no" issue (only "lost")
but was it ?

7 millions search results



Security of Data on Public Cloud Provider ? Privacy ?

Security != Privacy

You
can US Government ask Microsoft Azure to force get some data ?

ChatGPT
Yes, under certain circumstances and with proper legal authority, such as a warrant or subpoena, the US government can request data from Microsoft Azure or any other cloud service provider. Requests for data must comply with applicable laws and regulations, including those related to privacy and data protection. Microsoft and other technology companies typically have processes in place to handle such requests in accordance with the law.

You
can NSA use the USA Patriot Act to ask Microsoft Azure some data ?

ChatGPT
Yes, under certain circumstances, the NSA can use the authorities granted by the USA PATRIOT Act to request data from companies like Microsoft Azure. The PATRIOT Act expanded the government's surveillance powers, including the ability to issue National Security Letters (NSLs) to obtain various types of records and data, including electronic communications, from businesses and organizations, subject to certain legal requirements and oversight. However, the specifics of how and when these powers are utilized would depend on the specific circumstances and legal procedures involved.

Lows to restrict/allow Data on Public Cloud

You

laws to restrict/allows to store data on public cloud ?

ChatGPT

Laws and regulations regarding the storage of data on public clouds vary depending on the country or region. Here are some common considerations:

1. **Data Protection Laws**: Many countries have data protection laws that govern how personal data can be collected, processed, and stored. These laws often require consent for data storage and impose requirements for data security and confidentiality.
2. **Cross-Border Data Transfer Laws**: Some jurisdictions have regulations restricting the transfer of data across borders. If a public cloud provider stores data in a different country or region, they must comply with these regulations.
3. **Industry-Specific Regulations**: Certain industries, such as healthcare and finance, have specific regulations governing the storage and processing of data. These regulations often have strict requirements for data security and privacy.
4. **Government Access Laws**: Governments may have laws granting them access to data stored on public clouds for national security or law enforcement purposes. These laws often come with specific legal procedures and oversight mechanisms.
5. **Cloud Service Provider Terms of Service**: Public cloud providers typically have terms of service that outline how they handle customer data. These terms may include commitments to data security, privacy, and compliance with applicable laws and regulations.

It's essential for organizations considering storing data on public clouds to understand and comply with relevant laws and regulations to ensure data protection, privacy, and legal compliance. Consulting with legal experts knowledgeable about data protection and privacy laws in the relevant jurisdictions is advisable.

Data Across Border

How to practically check ?

Choose your "Region"

Trust

Example: Bank in Europe on Azure
=> choose Azure West-Europe, or Azure North Europe, ..

How to be sure "NO data" is sent to US ?

Laws ..

You

give sample for financial (banks) ?

ChatGPT

Certainly! Here are some examples of laws and regulations that may apply to banks and financial institutions regarding the storage of data on public clouds:

1. **Gramm-Leach-Bliley Act (GLBA)**: In the United States, the GLBA requires financial institutions to protect the privacy and security of customers' personal financial information. While the GLBA does not specifically address cloud computing, it mandates that banks implement appropriate safeguards to protect customer data, which would include ensuring the security of data stored on public clouds.
2. **Payment Card Industry Data Security Standard (PCI DSS)**: PCI DSS is a global standard that applies to organizations that handle payment card data. Financial institutions, including banks, must comply with PCI DSS requirements when storing payment card data on public clouds to ensure the security of sensitive financial information.
3. **European Union's General Data Protection Regulation (GDPR)**: GDPR applies to banks and financial institutions that process personal data of individuals in the European Union (EU), regardless of where the institution is located. GDPR imposes strict requirements for the protection of personal data, including requirements for data storage and transfer to third parties such as public cloud providers.
4. **Financial Industry Regulatory Authority (FINRA) Rules**: In the United States, FINRA regulates broker-dealers and requires them to maintain the confidentiality, integrity, and availability of customer data. While FINRA does not explicitly address cloud computing, financial institutions

Fears, Customers, ..
Data Classification C0,C1,C2
Eligibility to Cloud

Most companies classify their data with classification levels

example:
C0 = public data (ex: official CAC40 closing price yesterday)
C1 = internal data (ex: organigram of employees
C2 = private data (ex: trade with client, password)
C3 = ultra confidential data

example of program to cloud:
fears => up to C1
trust => C2

Encrypt Files vs Encrypt FileSystem

.exe

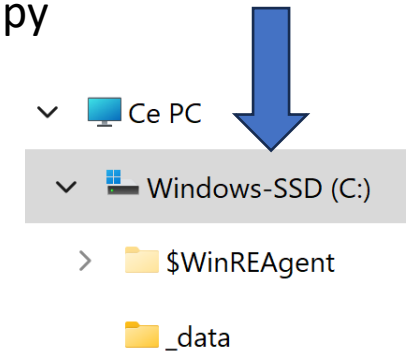
tmp copy
used by App

clear text
clear text

explicit encrypt
$ openssl enc file.txt ..

SALT_x$
%1!4xA6
9érP5°w

explicit copy







explicit decrypt
$ openssl enc file.txt ..

.exe

clear text
clear text

System call "write()"

System call "read()"

write
=> on the fly encrypt

read
=> on the fly decrypt



<opaque>
01010101

Mounting Encrypted FileSystem ex on Linux : LUKS (Linux Unified Key Setup)

You

in linux, how to mount an enncrypted disk filesystem ?

ChatGPT

To mount an encrypted disk filesystem in Linux, you typically follow these steps:

1. **Install Necessary Packages**: Ensure that the required packages for handling encrypted filesystems are installed. For LUKS (Linux Unified Key Setup) encrypted disks, you'll need the `cryptsetup` package.
2. **Open the Encrypted Disk**: Use the `cryptsetup` command to open the encrypted disk. Replace `/dev/sdX` with the path to your encrypted disk and `encrypted_volume` with the name you want to assign to the decrypted volume.

```bash
sudo cryptsetup luksOpen /dev/sdX encrypted_volume
```

   You'll be prompted to enter the passphrase or key to unlock the encrypted disk.

3. **Create a Mount Point**: Create a directory where you want to mount the decrypted filesystem.

```bash
sudo mkdir /mnt/encrypted
```

4. **Mount the Decrypted Filesystem**: Use the `mount` command to mount the decrypted filesystem to the mount point.

```bash
sudo mount /dev/mapper/encrypted_volume /mnt/encrypted
```

   Replace `/dev/mapper/encrypted_volume` with the decrypted device name obtained after running the `cryptsetup luksOpen` command.

5. **Access the Decrypted Files**: You can now access the files on the decrypted filesystem by navigating to the mount point (`/mnt/encrypted` in this example).
6. **Unmount and Close the Encrypted Disk**: After you're done working with the decrypted filesystem, unmount it and close the encrypted disk.

```bash
sudo umount /mnt/encrypted
sudo cryptsetup luksClose encrypted_volume
```

on Windows

You

and on windows ?

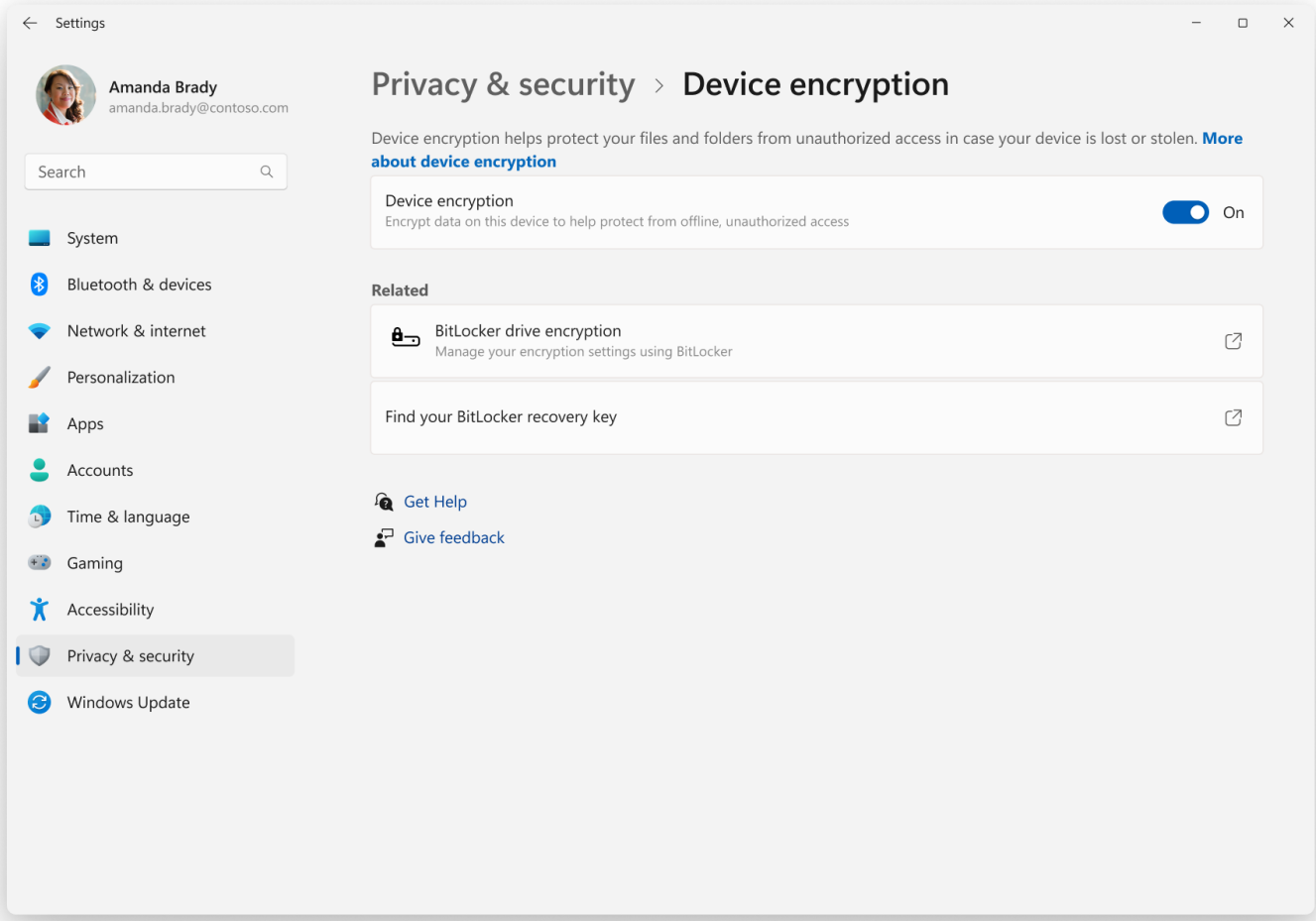
ChatGPT

In Windows, encryption of disks is commonly managed using BitLocker, which provides full disk encryption for Windows operating systems. Here's how encryption keys and passphrases are configured and managed in BitLocker:

1. **During Encryption Setup**: When you encrypt a disk using BitLocker, you have the option to specify either a passphrase or use a Trusted Platform Module (TPM) to store the encryption key. If you choose to use a passphrase, you'll be prompted to enter it during the encryption setup process.
2. **Stored Encryption Key**: If you're using a TPM, BitLocker automatically encrypts the disk's encryption key and stores it in the TPM. This key is used to encrypt and decrypt the data on the disk. If you're using a passphrase, BitLocker stores a copy of the encryption key encrypted with the passphrase in the disk's header.
3. **Prompted for Passphrase during Boot**: When you start or restart your computer, BitLocker prompts you to enter the passphrase or PIN to unlock the encrypted disk. If you're using a TPM, this step might be transparent to you, as BitLocker automatically retrieves the encryption key from the TPM.
4. **Encryption Key Management**: BitLocker manages the encryption key and interacts with the TPM or passphrase to unlock the encrypted disk during boot. BitLocker also provides tools for managing recovery keys, which are used to unlock the disk in case you forget your passphrase or encounter other issues during boot.
5. **Recovery Options**: BitLocker offers several recovery options in case you forget your passphrase or encounter problems during boot. These include using a recovery key, which is a numeric or alphanumeric code generated during encryption setup, or using a recovery password, which is a 48-digit numerical password displayed by BitLocker during setup.

In summary, on Windows systems using BitLocker, encryption keys and passphrases are configured during encryption setup and stored securely either in the TPM or encrypted with the passphrase in the disk's header. During boot, you're prompted to enter the passphrase or PIN to unlock the encrypted disk and access the data.

Windows BitLocker



Pseudonimisation / Anonymisation

You
what is the difference between pseudonynisation and anonymisation

ChatGPT
Pseudonymization and anonymization are both techniques used to protect the privacy of data, but they operate differently and offer varying levels of privacy protection:

1. **Pseudonymization**:
   - Pseudonymization involves replacing identifiable data with a pseudonym, which is a unique identifier or code. This process allows for the data to be associated with a substitute value rather than directly identifying information.
   - Pseudonymization doesn't completely remove the identifiability of data; it just makes it less directly identifiable. If there is a way to link the pseudonym back to the original identity, it's still possible to re-identify the individual.
   - Pseudonymization is often reversible, meaning that there is a mechanism to re-associate the pseudonyms with the original identifiers if needed.
   - Example: Replacing a person's name with a unique identifier in a database.
2. **Anonymization**:
   - Anonymization involves the irreversible removal or obscuring of identifiable information from data, making it impossible to link the data back to the original individual.
   - Anonymization typically involves more extensive transformations of data than pseudonymization, and it aims to provide a higher level of privacy protection by making re-identification practically impossible.
   - Anonymization techniques may include removing direct identifiers, generalizing data, or aggregating data to a level where individual identities cannot be discerned.
   - Once data is anonymized, it cannot be reversed to its original state to retrieve identifiable information.
   - Example: Removing all personally identifiable information (PII) such as names, addresses, and social security numbers from a dataset, and aggregating data to a level where individual contributions are indistinguishable.

Data Access Authorization

File System Permission
POSIX (umask) & ACL (Access-Control List)

File-system permissions

16 languages

Article Talk Read Edit View history Tools

From Wikipedia, the free encyclopedia

Most file systems include attributes of files and directories that control the ability of users to read, change, navigate, and execute the contents of the file system. In some cases, menu options or functions may be made visible or hidden depending on a user's permission level; this kind of user interface is referred to as permission-driven.

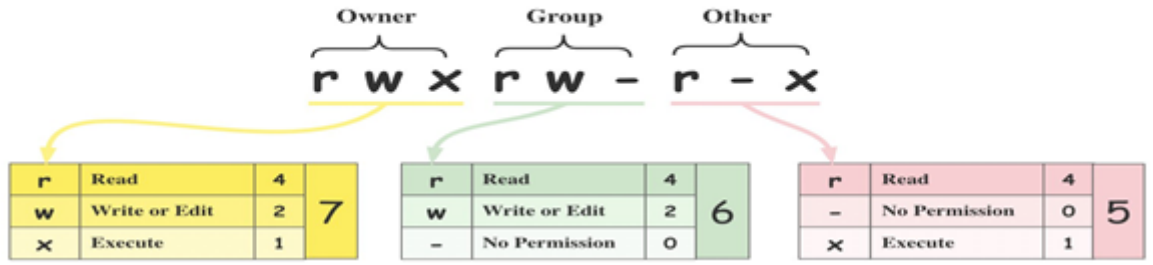
Two types of permissions are widely available: POSIX file system permissions and access-control lists (ACLs) which are capable of more specific control.

$ chmod go-rwx

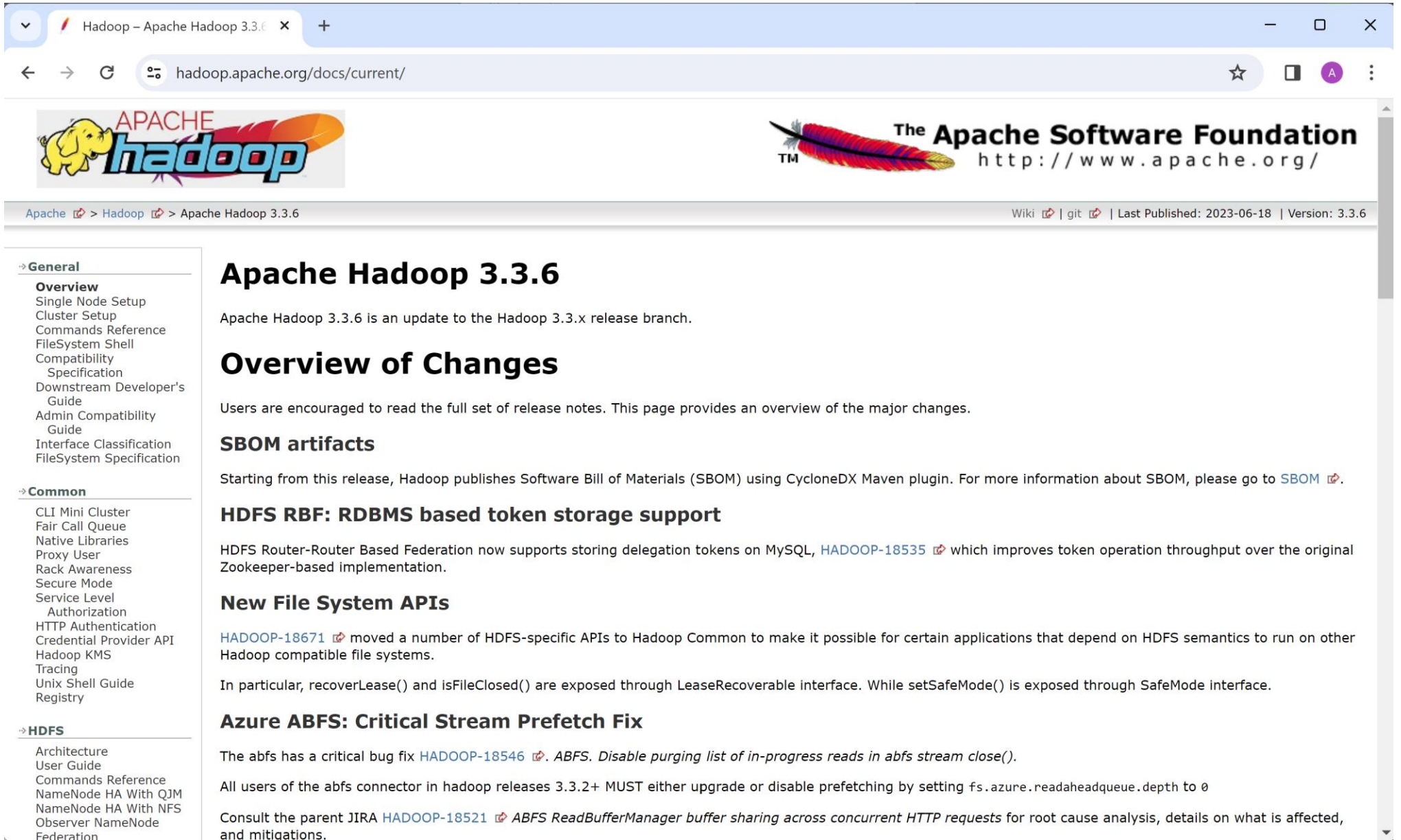
Linux File Permissions

blog.bytebytego.com

Binary	Octal	String Representation	Permissions
000	0 (0+0+0)	---	No Permission
001	1 (0+0+1)	--x	Execute
010	2 (0+2+0)	-w-	Write
011	3 (0+2+1)	-wx	Write + Execute
100	4 (4+0+0)	r--	Read
101	5 (4+0+1)	r-x	Read + Execute
110	6 (4+2+0)	rw-	Read + Write
111	7 (4+2+1)	rwx	Read + Write + Execute



https://hadoop.apache.org



HDFS
(Hadoop Distributed File System)

Hadoop KMS
Tracing
Unix Shell Guide
Registry

HDFS

Architecture
User Guide
Commands Reference
NameNode HA With QJM
NameNode HA With NFS
Observer NameNode
Federation
ViewFs
ViewFsOverloadScheme
Snapshots
Edits Viewer
Image Viewer
Permissions and HDFS
Quotas and HDFS
libhdfs (C API)
WebHDFS (REST API)
HttpFS
Short Circuit Local Reads
Centralized Cache Management
NFS Gateway
Rolling Upgrade
Extended Attributes
Transparent Encryption
Multihoming
Storage Policies
Memory Storage Support
Synthetic Load Generator
Erasure Coding
Disk Balancer
Upgrade Domain
DataNode Admin
Router Federation
Provided Storage

MapReduce

Tutorial

Notice Hadoop != HDFS

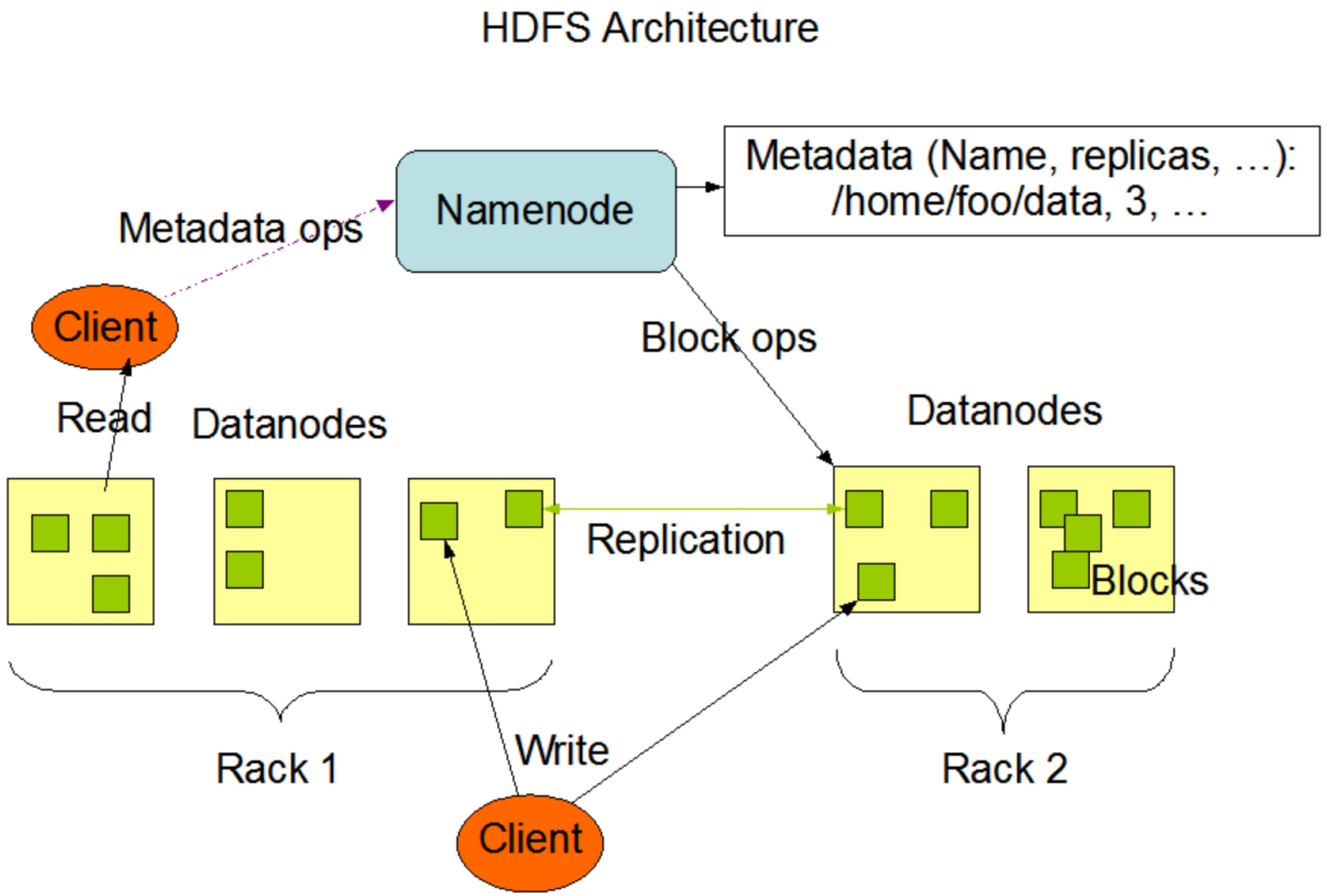
HDFS = sub-project of Hadoop (inpired for Google BigTable - File)

Hadoop File System = Java API (abstract class)

HDFS Architecture

NameNode and DataNodes

HDFS has a master/slave architecture. An HDFS cluster consists of a single NameNode, a master server that manages the file system namespace and DataNodes, usually one per node in the cluster, which manage storage attached to the nodes that they run on. HDFS exposes a file system namespac one or more blocks and these blocks are stored in a set of DataNodes. The NameNode executes file system namespace operations like opening, closin blocks to DataNodes. The DataNodes are responsible for serving read and write requests from the file system's clients. The DataNodes also perform b NameNode.



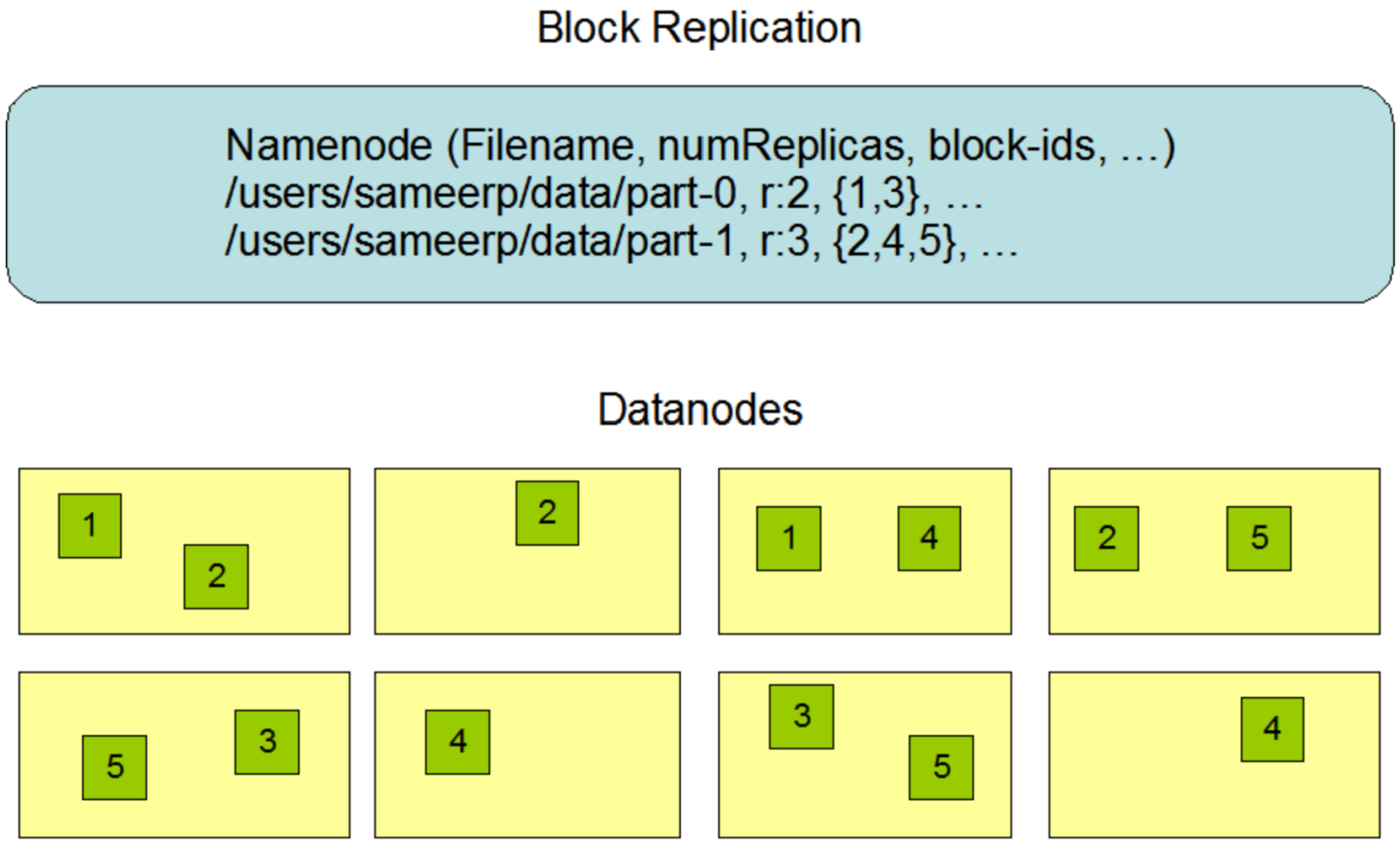
HDFS Scalability (DataNodes)

There can be >= Thousands of DataNodes

But always 2 NameNodes ... The bottle-neck of HDFS Scalability

all Files have their metadata in-memory on the NameNode
(~ 4k per file .... 100 Million Files => require ~400 Go of RAM)

HDFS Block Replication (RF = ReplicationFactor = 3)

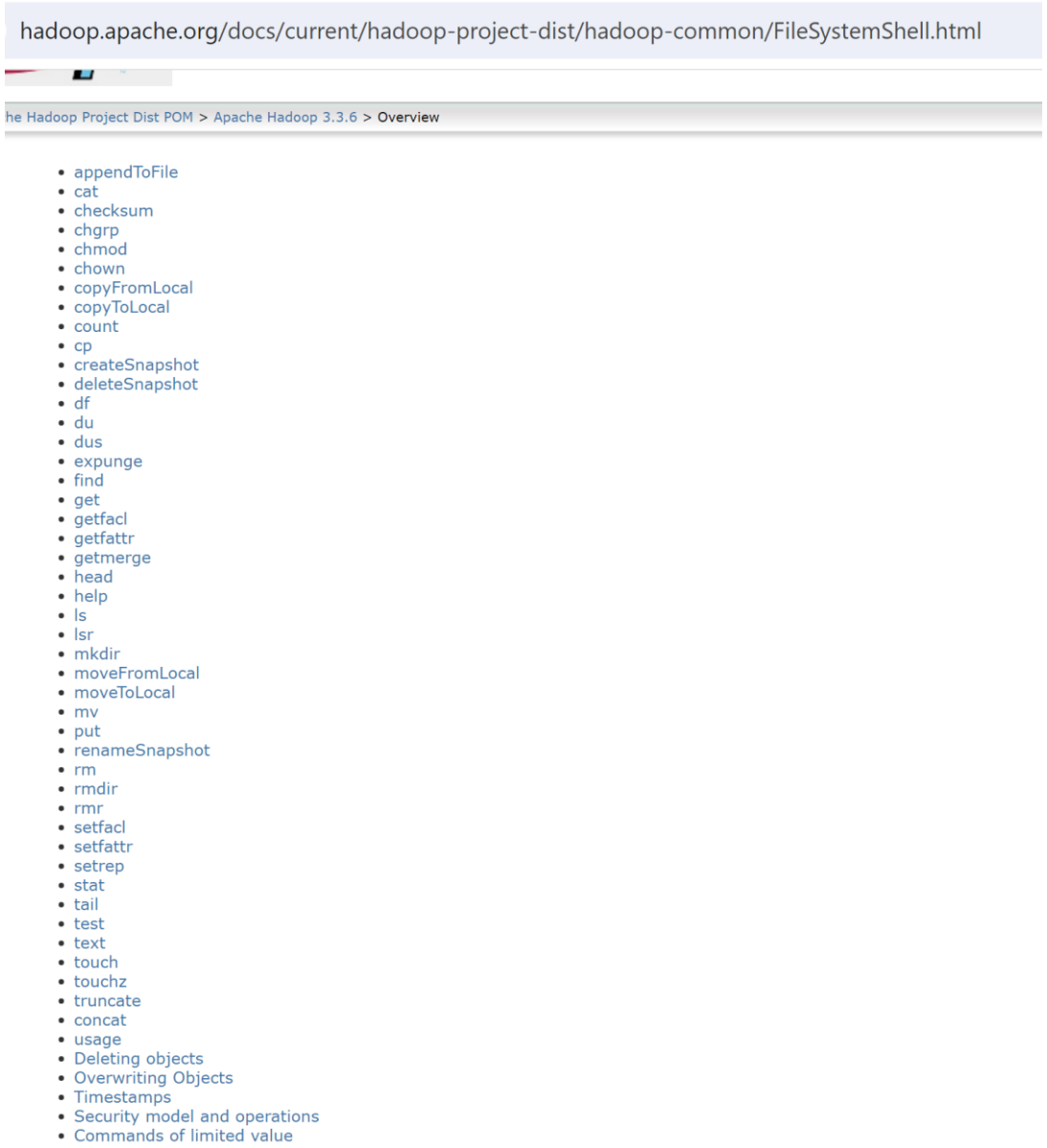


HDFS Client (Interfaces, Libraries, Shell)

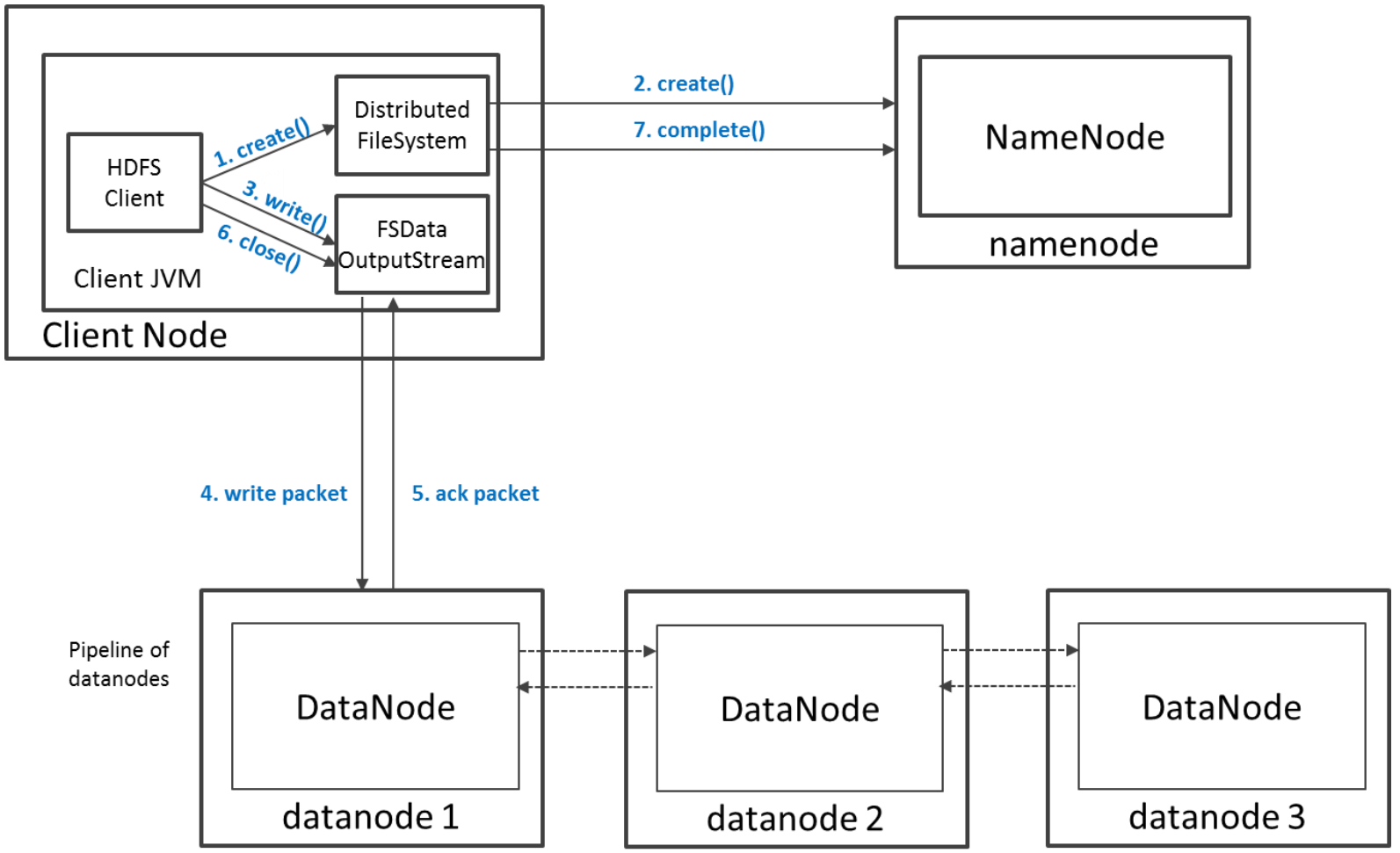
simplest: Shell Commands $ hdfs dfs <<command>>
most used: Java API (DistributedFileSystem)

others:
mountable Fuse
C wrapper (libhdfs)
HTTP protocol
WebDAV protocol
NFSGateway

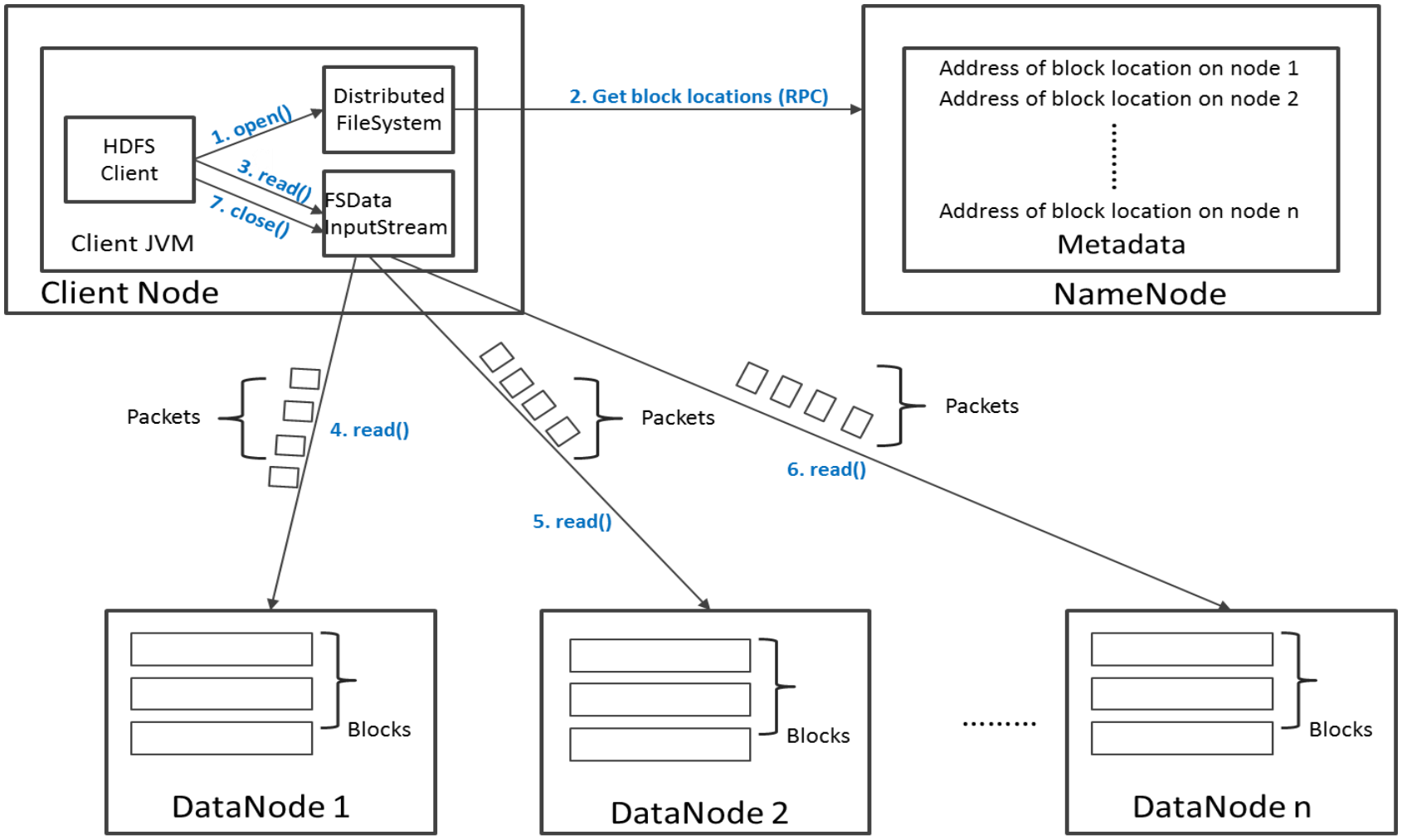
$ hdfs dfs -<<command>>
(= hadoop fs -<<command>>)



HDFS Write Operations



HDFS Read Operations



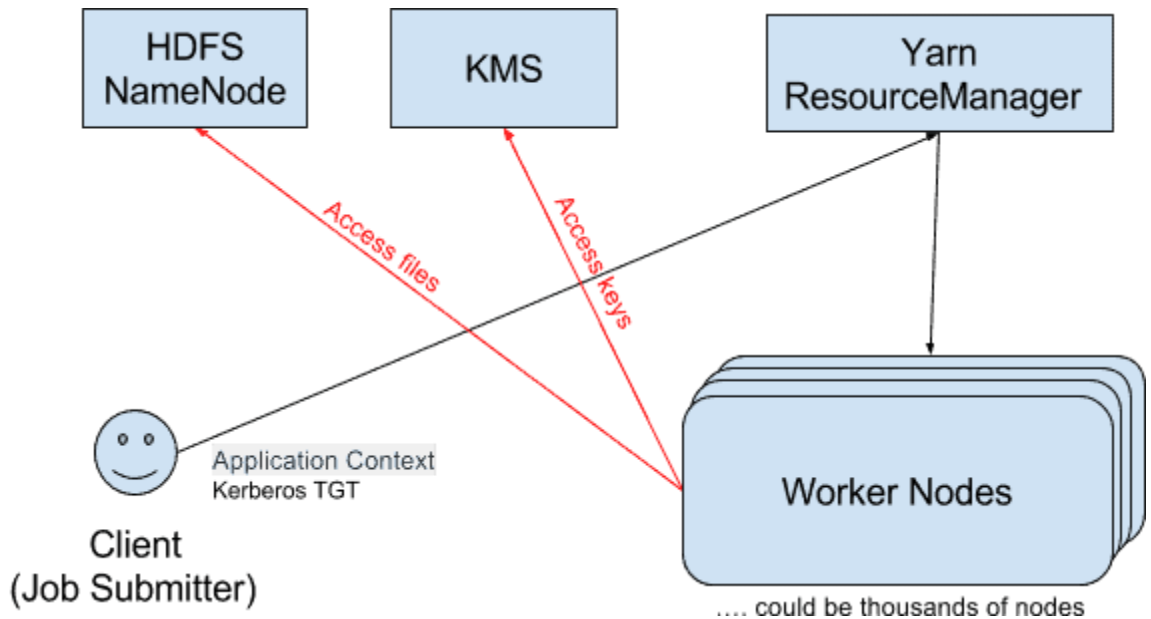
HDFS Authentication

2 possible modes :

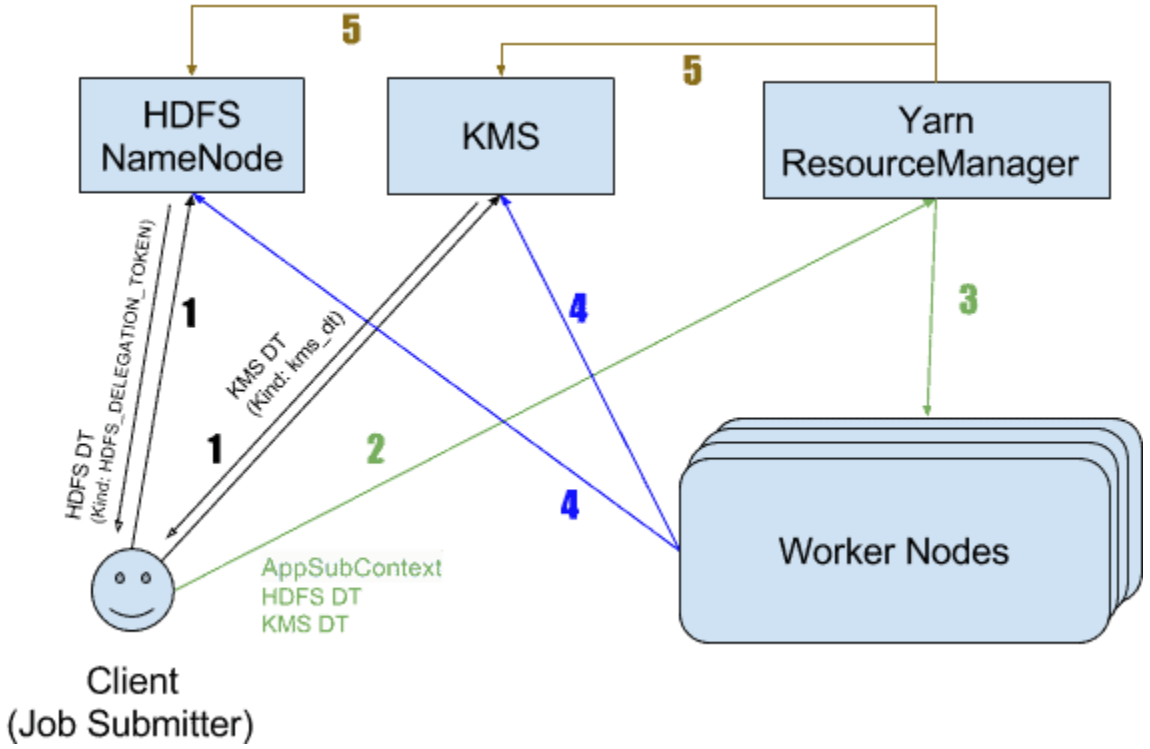
Simple –using OS username (unsecure?) to determine hadoop identity

Kerberos – authentication using kerberos ticket

Set by hadoop.security.authentication=simple|kerberos


HDFS
NameNode
KMS
Yarn
ResourceManager
Access files
Access keys
Application Context
Kerberos TGT
Client
(Job Submitter)
Worker Nodes
.... could be thousands of nodes

Hadoop Delegation Token



DelegationToken renew/max-lifetime

Property	Configuration Name in HDFS	Default Value in HDFS	Configuration Name in KMS	Default Value in KMS
Renew Interval	dfs.namenode.delegation.token.renew-interval	86400000 (1 day)	hadoop.kms.authentication.delegation-token.renew-interval.sec	86400 (1 day)
Max Lifetime	dfs.namenode.delegation.token.max-lifetime	604800000 (7 days)	hadoop.kms.authentication.delegation-token.max-lifetime.sec	604800 (7 days)

HDFS Authorizations : POSIX, ACL (checked by NameNode)

Standard POSIX Permissions for File and Directory
read (r), write (w), and execute (x) permissions
also has an owner, group and mode

enabled by default (dfs.permissions.enabled=true)

ACL = Access Control List
For more fine-grained permissions (!= owner user or group)

enabled by dfs.namenode.acls.enabled=true

hdfs POSIX Permissions

Changes to the File System API

All methods that use a path parameter will throw `AccessControlException` if permission checking fails.

New methods:

- `public FSDataOutputStream create(Path f, FsPermission permission, boolean overwrite, int bufferSize, short replication,`
- `public boolean mkdirs(Path f, FsPermission permission) throws IOException;`
- `public void setPermission(Path p, FsPermission permission) throws IOException;`
- `public void setOwner(Path p, String username, String groupname) throws IOException;`
- `public FileStatus getFileStatus(Path f) throws IOException;`will additionally return the user, group and mode associated with

The mode of a new file or directory is restricted by the umask set as a configuration parameter. When the existing `create(path, …)` met `^umask`. When the new `create(path, permission, …)` method (with the permission parameter P) is used, the mode of the new file is `P &` method (without the permission parameter), the mode of the new directory is `0777 & ^umask`. When the new `mkdirs(path, permission)` `^umask & 0777`.

Changes to the Application Shell

New operations:

- `chmod [-R] mode file ...`

  Only the owner of a file or the super-user is permitted to change the mode of a file.

- `chgrp [-R] group file ...`

  The user invoking chgrp must belong to the specified group and be the owner of the file, or be the super-user.

- `chown [-R] [owner][:[group]] file ...`

  The owner of a file may only be altered by a super-user.

- `ls file ...`

- `lsr file ...`

  The output is reformatted to display the owner, group and mode.

hdfs dfs -chgrp / hdfs dfs -chown
hdfs dfs -chmod

chgrp

Usage: `hadoop fs -chgrp [-R] GROUP URI [URI ...]`

Change group association of files. The user must be the owner of files, or else a super-user. Additional information is in the Permissions Guide.

Options

- The -R option will make the change recursively through the directory structure.

chmod

Usage: `hadoop fs -chmod [-R] <MODE[,MODE]... | OCTALMODE> URI [URI ...]`

Change the permissions of files. With -R, make the change recursively through the directory structure. The user must be the owner of the file, or else a super-user. Additional information is in the Permissions Guide.

Options

- The -R option will make the change recursively through the directory structure.

chown

Usage: `hadoop fs -chown [-R] [OWNER][:[GROUP]] URI [URI ]`

Change the owner of files. The user must be a super-user. Additional information is in the Permissions Guide.

Options

- The -R option will make the change recursively through the directory structure.

hdfs ACLs

ACLs (Access Control Lists)

In addition to the traditional POSIX permissions model, HDFS also supports POSIX ACLs (Access Control Lists). ACL
of users and groups. An ACL provides a way to set different permissions for specific named users or named groups,

By default, support for ACLs is enabled, and the NameNode allows creation of ACLs. To disable support for ACLs, se

An ACL consists of a set of ACL entries. Each ACL entry names a specific user or group and grants or denies read, v

```
user::rw-
user:bruce:rwx                  #effective:r--
group::r-x                      #effective:r--
group:sales:rwx                 #effective:r--
mask::r--
other::r--
```

ACL (default) mask

Every ACL must have a mask. If the user doesn't supply a mask while setting an ACL, ther

Running `chmod` on a file that has an ACL actually changes the permissions of the mask. Sin
group entry and possibly missing other extended ACL entries.

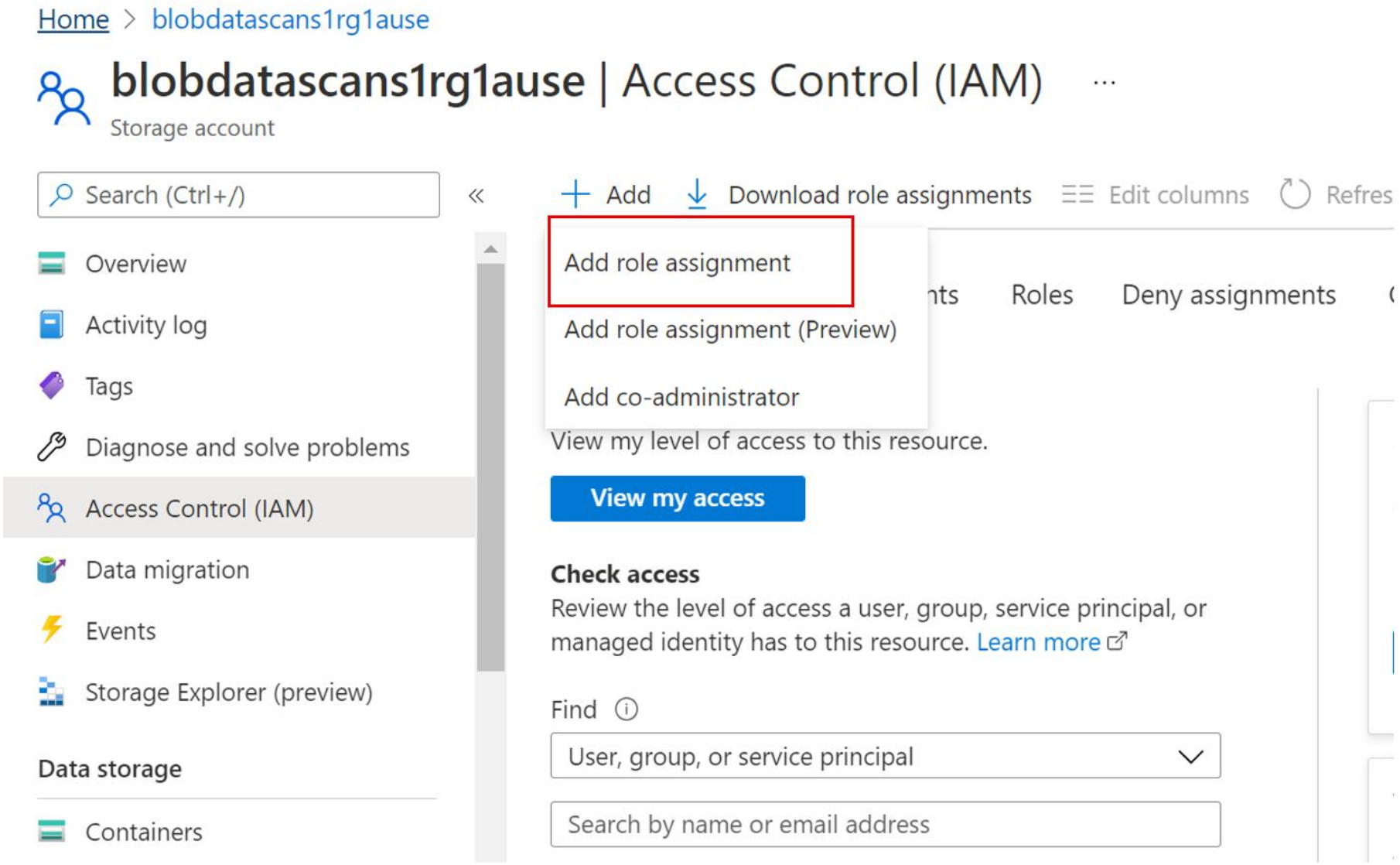
The model also differentiates between an "access ACL", which defines the rules to enforce
automatically during creation. For example:

```
user::rwx
group::r-x
other::r-x
default:user::rwx
default:user:bruce:rwx          #effective:r-x
default:group::r-x
default:group:sales:rwx         #effective:r-x
default:mask::r-x
default:other::r-x
```

Data on Microsoft Azure

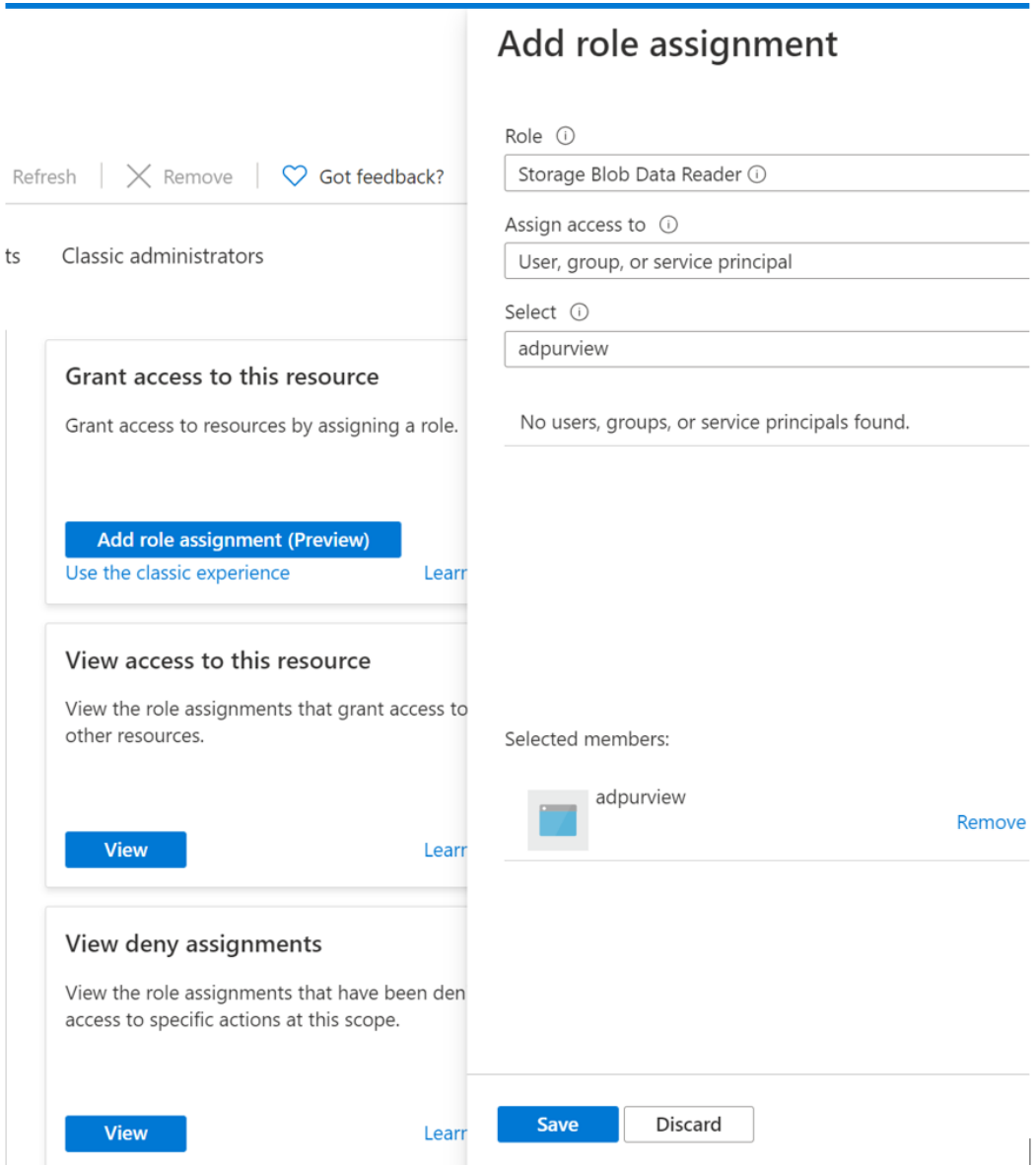
Azure Datalake FileSystem Storage
abfss://

Azure Storage - Access Control (IAM)

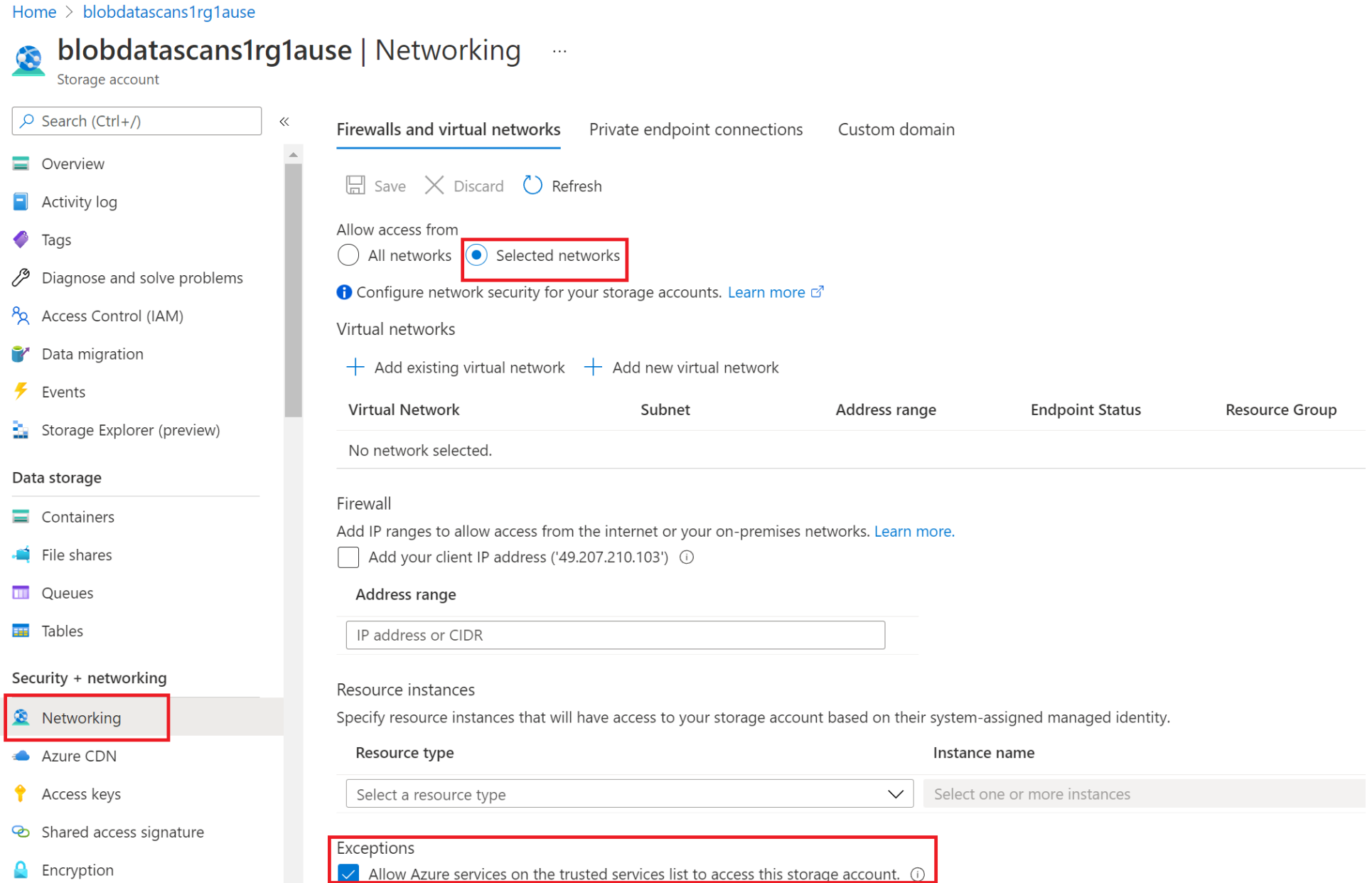


Role: Storage Blob Data Owner

Role: Storage Blob Data Reader



Access from VNet



Data Governance

Roles of the CDO (Chief Data Officer) ?

You

wht are the roles of a Chief Data Officer,

ChatGPT

The Chief Data Officer (CDO) plays a crucial role in modern organizations, particularly those that rely heavily on data-driven decision-making processes. Here are some key roles and responsibilities of a Chief Data Officer:

1. **Data Strategy Development**: The CDO is responsible for developing and implementing a comprehensive data strategy aligned with the organization's overall business objectives. This includes defining data governance policies, data quality standards, data architecture, and data management practices.
2. **Data Governance**: Ensuring that the organization's data assets are managed effectively and securely is a core responsibility of the CDO. This involves establishing data governance frameworks, defining data ownership and stewardship, and enforcing compliance with data regulations and policies.
3. **Data Quality Management**: Maintaining high standards of data quality is essential for accurate and reliable decision-making. The CDO oversees initiatives to assess, improve, and maintain the quality of the organization's data assets, including data cleansing, validation, and enrichment processes.
4. **Data Analytics and Insights**: Leveraging data analytics to derive actionable insights is a key function of the CDO. This involves working closely with data scientists, analysts, and business stakeholders to identify business opportunities, develop predictive models, and drive data-driven decision-making across the organization.
5. **Data Privacy and Security**: With increasing concerns about data privacy and security, the CDO plays a crucial role in ensuring that data assets are protected from unauthorized access, breaches, and misuse. This involves implementing robust security measures, overseeing compliance with data privacy regulations (such as GDPR or CCPA), and managing risks related to data security.
6. **Data Innovation and Monetization**: The CDO is responsible for identifying opportunities to monetize data assets and drive innovation through data-driven products and services. This may involve exploring new data sources, developing data partnerships, and creating new revenue streams based on data insights.
7. **Stakeholder Engagement and Collaboration**: Effective communication and collaboration with business leaders, IT teams, data professionals, and other stakeholders are essential for the success of data initiatives. The CDO serves as a liaison between different departments, aligning data priorities with business objectives and fostering a data-driven culture within the organization.

Overall, the Chief Data Officer plays a critical role in driving value from data assets, enabling informed decision-making, and fostering a culture of data-driven innovation and excellence within the organization.

Data Governance Challenges: Data Consumer

Challenges for data consumers

Traditionally, discovering enterprise data sources has been an organic process based on communal knowledge. For companies that want the most value from their information assets, this approach presents many challenges:

- Because there's no central location to register data sources, users might be unaware of a data source unless they come into contact with it as part of another process.
- Unless users know the location of a data source, they can't connect to the data by using a client application. Data-consumption experiences require users to know the connection string or path.
- The intended use of the data is hidden to users unless they know the location of a data source's documentation. Data sources and documentation might live in several places and be consumed through different kinds of experiences.
- If users have questions about an information asset, they must locate the expert, or team responsible for that data and engage them offline. There's no explicit connection between the data and the experts that understand the data's context.
- Unless users understand the process for requesting access to the data source, discovering the data source and its documentation won't help them access the data.

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Challenges for data producers

Although data consumers face the previously mentioned challenges, users who are responsible for producing and maintaining information assets face challenges of their own:

- Annotating data sources with descriptive metadata is often a lost effort. Client applications typically ignore descriptions that are stored in the data source.
- Creating documentation for data sources can be difficult and it's an ongoing responsibility to keep documentation in sync with data sources. Users might not trust documentation that's perceived as being out of date.
- Creating and maintaining documentation for data sources is complex and time-consuming. Making that documentation readily available to everyone who uses the data source can be even more so.
- Restricting access to data sources and ensuring that data consumers know how to request access is an ongoing challenge.

When such challenges are combined, they present a significant barrier for companies that want to encourage and promote the use and understanding of enterprise data.

Data Governance Challenges: Data Consumer

Challenges for security administrators

Users who are responsible for ensuring the security of their organization's data may have any of the challenges listed above as data consumers and producers, and the following extra challenges:

- An organization's data is constantly growing and being stored and shared in new directions. The task of discovering, protecting, and governing your sensitive data is one that never ends. You need to ensure that your organization's content is being shared with the correct people, applications, and with the correct permissions.
- Understanding the risk levels in your organization's data requires diving deep into your content, looking for keywords, RegEx patterns, and sensitive data types. For example, sensitive data types might include Credit Card numbers, Social Security numbers or Bank Account numbers. You must constantly monitor all data sources for sensitive content, as even the smallest amount of data loss can be critical to your organization.
- Ensuring that your organization continues to comply with corporate security policies is a challenging task as your content grows and changes, and as those requirements and policies are updated for changing digital realities. Security administrators need to ensure data security in the quickest time possible.

European Digital Service Act

Digital Services Act

13 languages

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From Wikipedia, the free encyclopedia

The **Digital Services Act Regulation 2022** (EU) 2022/2065 ("DSA") is a regulation in EU law to update the Electronic Commerce Directive 2000 regarding illegal content, transparent advertising, and disinformation.[1][2] It was submitted along with the Digital Markets Act (DMA) by the European Commission to the European Parliament and the Council on 15 December 2020.[3][4] The DSA was prepared by the Executive Vice President of the European Commission for A Europe Fit for the Digital Age Margrethe Vestager and by the European Commissioner for Internal Market Thierry Breton, as members of the Von der Leyen Commission.[5]

On 22 April 2022, European policymakers reached an agreement on the Digital Services Act.[6][7] The European Parliament approved the DSA along with the Digital Markets Act on 5 July 2022.[8] On 4 October 2022, the European Council gave its final approval to the Regulation on a Digital Services Act.[9] It was published in the Official Journal of the European Union on 19 October 2022. Affected service providers will have until 1 January 2024 to comply with its provisions. Popular online platforms and search engines will need to comply with their obligations four months after they have been designated as such by the EU Commission.[8]

Regulation (EU) 2022/2065

European Union regulation	
Text with EEA relevance	
Title	Regulation on a Single Market For Digital Services
Made by	European Parliament and Council of the European Union
Journal reference	OJ L 277, 27.10.2022, p. 1–102
History	
Date made	19 October 2022
Preparative texts	
Commission proposal	COM/2020/825 final
Current legislation	

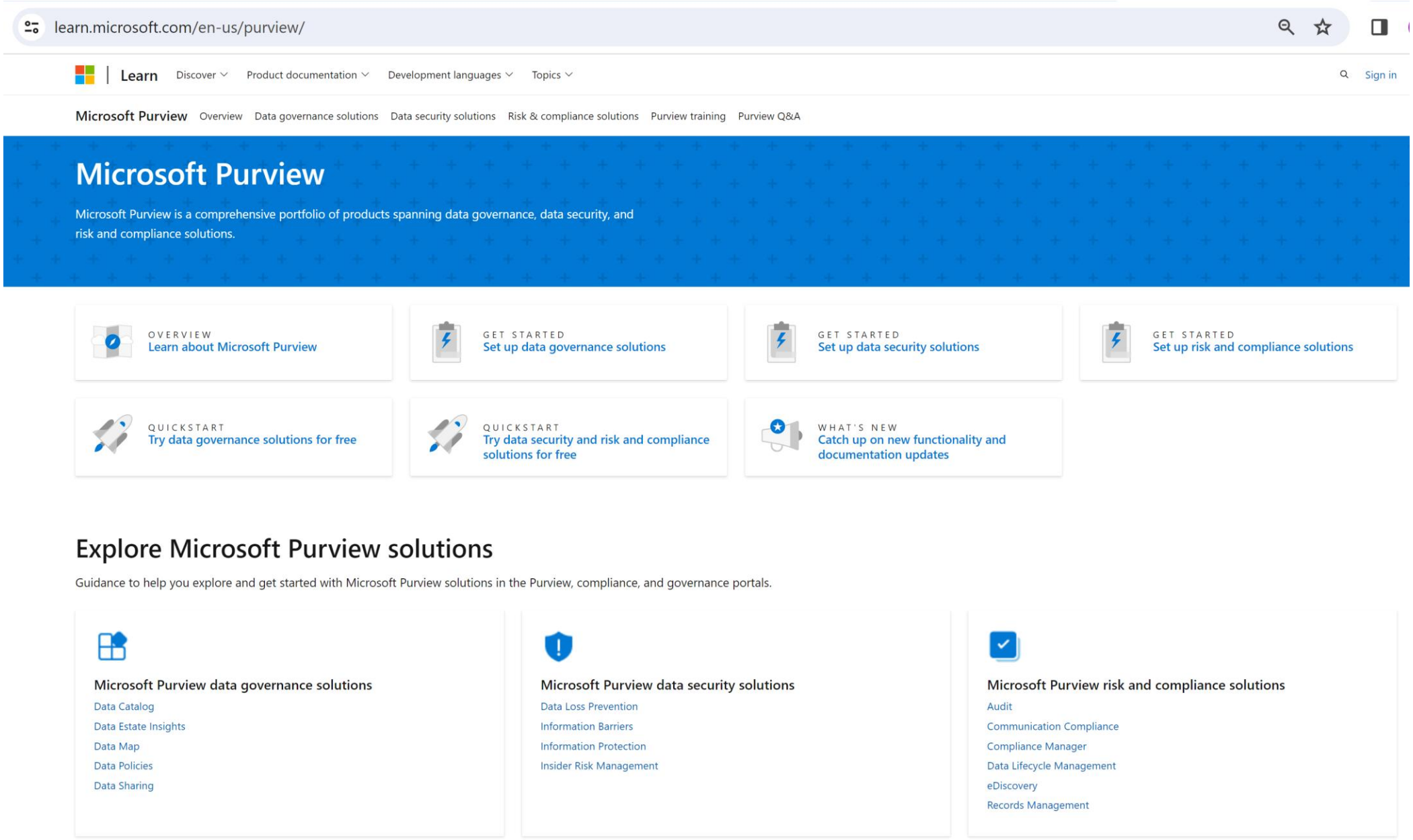
Objectives of the DSA [edit]

Ursula von der Leyen proposed a "new Digital Services Act", in her 2019 bid for the European Commission's presidency.[10]

The expressed purpose of the DSA is to update the European Union's legal framework for illegal content on intermediaries, in particular by modernising the e-Commerce Directive adopted in 2000. In doing so, the DSA aims to harmonise different national laws in the European Union that have emerged at national level to address illegal content.[1] Most prominent amongst these laws has been the German NetzDG, and similar laws in Austria ("Kommunikationsplattformen-Gesetz") and France ("Loi Avia"). With the adoption of the Digital Services Act at European level, those national laws would be overwritten and would have to be repealed.[11]

Microsoft PurView

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PurView

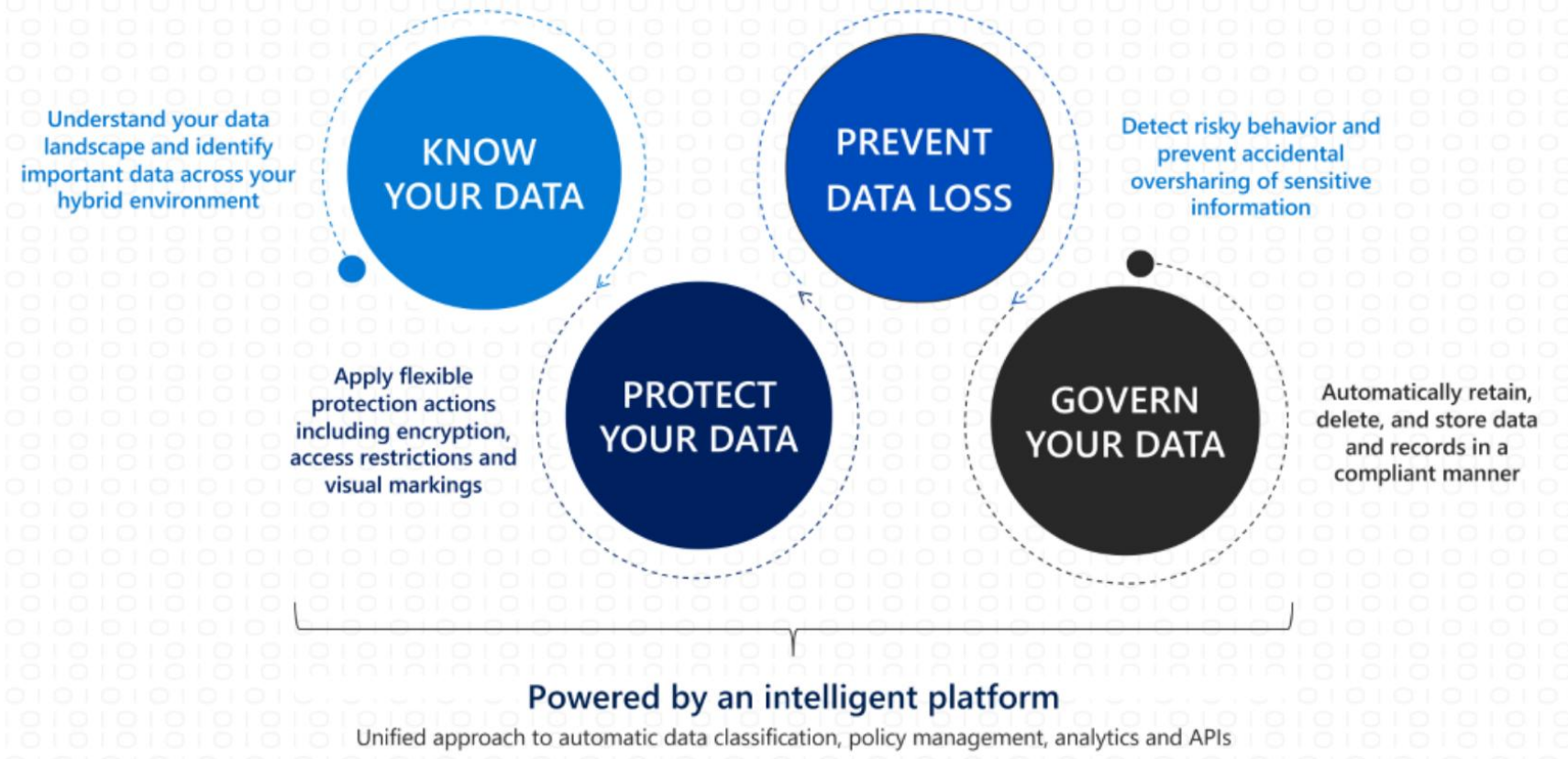
Microsoft Purview combines the former Azure Purview and Microsoft 365 compliance solutions and services together into a unified platform to help your organization:

- Gain visibility into data across your organization
- Safeguard and manage sensitive data across its lifecycle, wherever it lives
- Govern data seamlessly in new, comprehensive ways
- Manage critical data risks and regulatory requirements



Data security, governance, and compliance together

Data: Know -> Protect -> PreventLoss -> Govern



Data Loss Prevention (avoid sharing sensitive data, example: no copy/email)

Learn about data loss prevention

Article • 01/09/2024 • 5 contributors

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In this article

Before you begin
Licensing and Subscriptions
DLP is part of the larger Microsoft Purview offering
Protective actions of DLP policies
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Organizations have sensitive information under their control such as financial data, proprietary data, credit card numbers, health records, or social security numbers. To help protect this sensitive data and reduce the risk from oversharing, they need a way to help prevent their users from inappropriately sharing sensitive data with people who shouldn't have it. This practice is called data loss prevention (DLP).

In Microsoft Purview, you implement data loss prevention by defining and applying DLP policies. With a DLP policy, you can identify, monitor, and automatically protect sensitive items across:

- Microsoft 365 services such as Teams, Exchange, SharePoint, and OneDrive accounts
- Office applications such as Word, Excel, and PowerPoint
- Windows 10, Windows 11, and macOS (three latest released versions) endpoints
- non-Microsoft cloud apps
- on-premises file shares and on-premises SharePoint
- Power BI

Information Barriers: Chinese Walls

Information barriers

Article • 10/17/2023 • 2 contributors

Feedback

In this article

Configure information barriers
More information about information barriers

Microsoft 365 enables communication and collaboration across groups and organizations and supports ways to restrict communication and collaboration among specific groups of users when necessary. This requirement may include situations or scenarios where you want to restrict communication and collaboration between two groups to avoid a conflict of interest from occurring in your organization. This scenario may also include situations when you need to restrict communication and collaboration between certain people inside your organization to safeguard internal information.

Microsoft Purview Information Barriers (IB) is supported in Microsoft Teams, SharePoint Online, and OneDrive for Business. A compliance administrator or IB administrator can define policies to allow or prevent communications between groups of users in Microsoft Teams. Use IB policies for situations like these:

- User in the day trader group should not communicate or share files with the marketing team.
- Instructors in one school shouldn't be able to communicate or share files with students in another school in the same school district.